

Credit Supply Shocks and Firm Dynamics: Evidence From Brazil

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This paper

Role of **Financial friction** on Firm dynamics

- Testable mechanism: **Easing credit access** →
 - ① Extensive margin: ↑ entry of marginal firms? or with greater capability?
 - ② Intensive margin: ↑ growth among incumbent firms?
- Investigated a large-scale credit expansion for small and medium enterprises in Brazil (2003-2013): **Cartão BNDES credit line**
- Empirical findings: **Cartão-induced credit supply shock** have significant effects on firm turnover and entrant capability
 - ▶ 10 % increase in local credit supply → 6% increase in entry of new firms;
 - ▶ almost identical effects on firm exit → no change in overall formal-sector employment (at municipality level)
 - ▶ More frequent entry of firms that survive > two years, at larger scale, grow faster than those in municipalities with smaller credit expansions
 - ▶ Effects are more concentrated in municipalities with ex-ante weaker credit markets

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Outline

- 1 Background
- 2 Model
- 3 Identification
- 4 Results
- 5 Conclusion
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Description of Cartão program

- Brazil's National Development Bank (Banco Nacional de Desenvolvimento Econômico e Social or **BNDES**)
 - ▶ Legacy of industrial policy-based direct lending to large firms
 - ▶ Since 2002, targeted around 1/4 of total disbursements to small/medium enterprises (**SMEs**)
 - ▶ Cartão: piloted in March 2003, grew from 1 % of total BNDES lending to 12 % by 2013
- Cartão: rotating credit line for formal firms - input purchases from BNDES-accredited Brazilian suppliers
 - ▶ Cartão loans are smaller/more favorable terms than other prevailing credit market opportunities for SMEs
eg) annual interest rates 5 – 15% vs. other loans 25 – 30%
 - ▶ broad-limits pre-specified by BNDEs, intermediating banks set the actual terms (incl. collateral requirements, interest rates for the borrower, credit limits, repayment frequency)

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Description of Cartão program (2)

- Municipality-level summary statistics ▶ summary stat
- Heterogeneity across regions in deepening of credit markets ▶ evolution
▶ banking network
- Borrowing profiles of firms ▶ summary stat
- Possible spillover effect of the program ▶ borrowing

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Model (Bird's eye view)

- Model of **entrepreneurial choice** under **financial constraints** + Heterogeneous firms with monopolistic competition

(Evans and Jovanovic 1989) (Melitz 2003)

- M municipalities in the economy (differ in **capital market conditions**)
- Agents: a continuum L_m of households endowed with (z, ϕ)
 - z : personal wealth; ϕ : entrepreneurial idea
 - no uncertainty, joint distribution $F(\phi, z)$
 - Choice between becoming a worker and an entrepreneur

⇒ **Selection into entrepreneurs** will determine firm size and entry/exit
- Market
 - 1 Product Market: CES preference + monopolistic competition
 - 2 Labor Market: w_m clear the local market in municipality m
 L^s (workers) = L^d (**constrained entrepreneurs**) + L^d (unconstrained entrepreneurs)
 - 3 Capital rental market: borrowing rate = lending rate r

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Model - Agents

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$$0 \leq k \leq \lambda_m z = (\lambda_m - 1)z + z \quad (1)$$

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$$0 \leq k \leq \lambda_m z = (\lambda_m - 1)z + z \quad (1)$$

- If $z < k$: net borrower
- Equilibrium selection: no default i.e. $\pi + rz > 0$

Model - Product market

- Demand: CES + monopolistic competition

$$y(v) = Ap(v)^{-\sigma}, \quad A \equiv \frac{\sum_m w_m L_{\ell m} + \int_v \pi(v) dv + rZ}{p^{1-\sigma}} \quad (2)$$

- Firm's production function $q = f(k, \ell) = \phi k^\alpha \ell^{1-\alpha}$

$$\begin{aligned} x(v) &= p(v)y(v) = A^{\frac{1}{\sigma}} y(v)^{\frac{\sigma-1}{\sigma}} \\ \rightarrow x(k, \ell) &= A^{1/\sigma} [\phi k^\alpha \ell^{1-\alpha}]^{\frac{\sigma-1}{\sigma}} \end{aligned}$$

Model - Entrepreneurial choice

- Production (conditional on choosing entrepreneurship)

$$\max_{k \in [0, \lambda_m z], \ell} x(k, \ell) - rk - w_m \ell$$

- ▶ Interior solution ($k < \lambda_m z$) (financially unconstrained):
- ▶ Boundary solution ($k = \lambda_m z$) (financially constrained)

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$$\max_{k \in [0, \lambda_m z], \ell} x(k, \ell) - rk - w_m \ell$$

- Interior solution ($k < \lambda_m z$) (financially unconstrained):

$$\ell_m = \left(\frac{1-\alpha}{\mu} \right)^\sigma A \left[\phi \left(\frac{\alpha}{1-\alpha} \frac{w_m}{r} \right)^\alpha w_m^{-\mu} \right]^{\sigma-1} \quad (\text{Ua})$$

$$k_m = \left(\frac{\alpha}{\mu} \right)^\sigma A \left[\phi \left(\frac{1-\alpha}{\alpha} \frac{r}{w_m} \right)^{1-\alpha} r^{-\mu} \right]^{\sigma-1} \quad (\text{Ub})$$

$$\rightarrow \phi \leq D_m (\lambda_m)^{\mu-1} \cdot z^{\mu-1}; \quad D_m \equiv A^{-(\mu-1)} \mu^\mu r^\alpha w_m^{1-\alpha} \left(\frac{r}{\alpha} \right)^{\mu-1} / B \quad (5)$$

- Boundary solution ($k = \lambda_m z$) (financially constrained)

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$$\ell_{mz} = \left\{ \left(\frac{1 - \alpha}{\mu} \right)^\sigma A [\phi (\lambda_m z)^\alpha w_m^{-\mu}]^{\sigma-1} \right\}^{\frac{\mu/\sigma}{\alpha + \mu/\sigma}}. \quad (\text{Ca})$$

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- Employment as a proxy for size

$$\pi_m = \frac{\mu w_m}{\sigma(1-\alpha)} \ell_m \quad \text{and} \quad \pi_{mz} = \frac{[\mu - (1-\alpha)] w_m}{1-\alpha} \ell_{mz} - r \lambda_m z$$

$$\frac{\ell_{mz}}{\ell_m} = \left(\frac{\lambda_m z}{k_m} \right)^{\frac{\alpha}{\alpha + \mu/\sigma}} < 1 \quad \text{and} \quad \frac{\ell_{mz}/k_{mz}}{\ell_m/k_m} = \left(\frac{\lambda_m z}{k_m} \right)^{-\frac{\mu/\sigma}{\alpha + \mu/\sigma}} > 1$$

Model - Firm Entry and Exit

- Entrepreneurial net income

$$\pi_m(\phi, z; \lambda_m) = \begin{cases} \frac{\mu w_m}{\sigma(1-\alpha)} \ell_m(\phi) & \text{Unconstrained} \\ \frac{[\mu - (1-\alpha)] w_m}{1-\alpha} \ell_{mz}(\phi, z; \lambda_m) - r \lambda_m z & \text{Constrained} \end{cases}$$

Becomes entrepreneur iff $\pi_m(\phi, z; \lambda_m) \geq w_m$

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Becomes entrepreneur iff $\pi_m(\phi, z; \lambda_m) \geq w_m$

- **Unconstrained:**

$$\underline{\phi}_m = \mu \left(\frac{\sigma}{A} \right)^{\mu-1} \frac{r^\alpha w_m^{\mu-\alpha}}{B}$$

- **Constrained:**

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- Unconstrained:

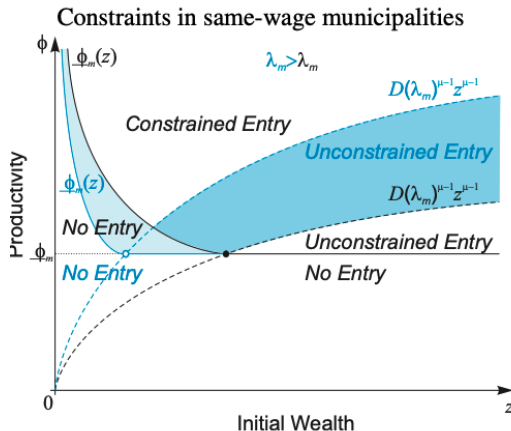
$$\phi_m = \mu \left(\frac{\sigma}{A} \right)^{\mu-1} \frac{r^\alpha w_m^{\mu-\alpha}}{B}$$

- Constrained:

$$\ell_{mz}(\phi_m(z), z; \lambda_m) - \frac{(1-\alpha)(r \lambda_m z + w_m)}{[\mu - (1-\alpha)] w_m} = 0$$

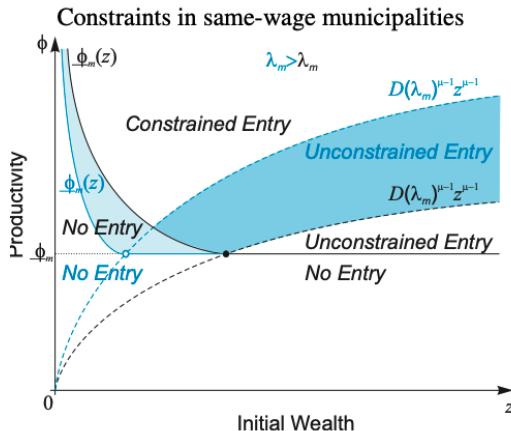
$$\left. \frac{d\phi_m(z)}{dz} \right|_{\pi(\phi, z; \lambda_m) = w_m} = -\frac{\phi}{z} \left[\alpha - (1-\alpha) \frac{r}{w_m} \lambda_m z \frac{1}{\ell_{mz}(\phi, z; \lambda_m)} \right] < 0.$$

Insight for growth mechanism under financial constraint



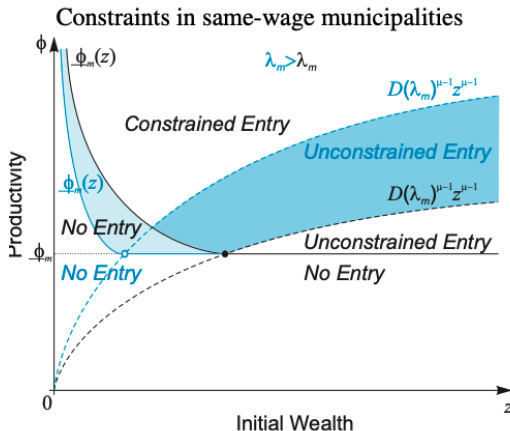
- (1) Growth rate of entrants
- (2) Entry at size

Insight for growth mechanism under financial constraint



- (1) Growth rate of entrants $\rightarrow \uparrow$ new entrants
- (2) Entry at size

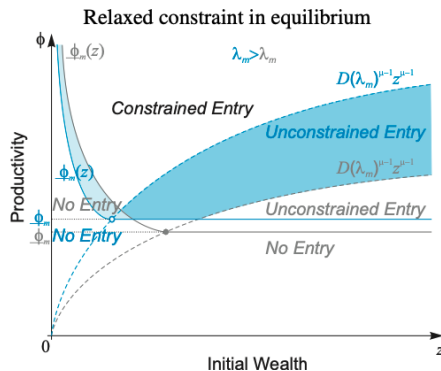
Insight for growth mechanism under financial constraint



- (1) Growth rate of entrants
- (2) Entry at size $\rightarrow$ ambiguous..needs empirical resolution

Relaxation of financial constraint

(Local) Equilibrium consequence of $\uparrow \lambda_m$



- $\uparrow w_m$ $\uparrow$ income for workers, $\uparrow$ product price $p(\nu)$
- higher labor costs/factor price $\downarrow$ firm optimal size, $\uparrow$ exit of firms
- Exact GE effects depend on the specification of $F(\phi, z)$

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Data: Firms and Finance

- **Credit Data:** Credit Information System Database (Sistema de Informações de Crédito or SCR)
 - ▶ transactions-level information on all loans ($> US\$2,700$ in December 2011) from 2003-11 and all loans ($> US\$300$ in December 2016) from 2012 to 2016
 - ▶ main variable of interest = value of disbursements to a given firm in a given year
- **Matched Employer-Employee Data:** Relação Anual de Informações Sociais (simply RAIS)
 - ▶ firm-level employment records by unique tax ID
 - ▶ comprise of all formal firms (≥ 1 employee)

Data - Firm entry/exit

(1) Measurement of Firm Entry (first yr of formal sector activity ≥ 1 employee)

- ① first year of appearance in RAIS data (extending back to 1986)
 - ▶ Concern: some firms may legally register but not yet pay workers and thereby enter the RAIS records)
- ② Define firm entry based on the longest reported tenure across all workers in a firm's first year in RAIS (to capture formal entry from prior informal sector)
 - ▶ could be still biased if original employees of the informal firm separated prior to the first yr of the firm's formalization

(2) Firm Exit: record exit in year t based on absence from RAIS for at least the next three years ($t+1$, $t+2$, $t+3$)

Empirical strategy: Isolating Credit Supply Shocks (1)

- Estimating Equations:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{carto}}) + \theta_{rt} + \varepsilon_{mrt} \quad (1)$$

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- Estimating Equations:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{carto}}) + \theta_{rt} + \varepsilon_{mrt} \quad (1)$$

- y_{mrt} : agg. measure of firm dynamics in municipality m in region r at time t

- $\ln(\text{credit}_{mrt}^{\text{carto}})$: log value of new Cartão loan approvals for firms operating in m in year t

Empirical strategy: Isolating Credit Supply Shocks (1)

- Estimating Equations:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{carto}}) + \theta_{rt} + \varepsilon_{mrt} \quad (1)$$

- Possible endogeneity concern: need to isolate changes in local Cartão **credit supply** orthogonal to changes in **credit demand**
- Unique institutional features: only certain banks could intermediate the program at certain points in time

Empirical strategy: Isolating Credit Supply Shocks (2)

- Estimating Equations:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{cartao}}) + \theta_{rt} + \varepsilon_{mrt} \quad (1)$$

- (1) Separate out bank-specific from municipality-specific in $\text{credit}_{mt}^{\text{Cartão}}$:

$$\Delta \ln(\text{credit}_{bmt}^{\text{cartao}}) = \alpha + S_{bt} + D_{mt} + \varepsilon_{bmt}$$

- (2) Construct IV for supply shock:

$$\widetilde{\text{shock}_{mt}^{\text{Cartão}}} = \sum_b \left(\frac{\text{credit}_{bm,t-1}^{\text{cartao}}}{\text{credit}_{m,t-1}^{\text{cartao}}} \right) \hat{S}_{bt}$$

Empirical strategy: Isolating Credit Supply Shocks (3)

- Estimating Equations:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{carto}}) + \theta_{rt} + \varepsilon_{mrt} \quad (1)$$

- Second-stage:

$$y_{mrt} = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{Cartão}}) + \delta \ln(\text{credit}_{mrt}^{\text{bndes}}) + \theta_{rt} + \varepsilon_{mrt}$$

- First-stage:

$$\ln \text{credit}_{mrt}^{\text{Cartão}} = \alpha^c + \phi^c \widetilde{\text{shock}_{mt}^{\text{Cartão}}} + \xi^c \widetilde{\text{shock}_{mt}^{\text{bndes}}} + \theta_{rt} + \nu_{mrt}^c$$

$$\ln \text{credit}_{mrt}^{\text{bndes}} = \alpha^{nc} + \phi^{nc} \widetilde{\text{shock}_{mt}^{\text{bndes}}} + \xi^{nc} \widetilde{\text{shock}_{mt}^{\text{Cartão}}} + \theta_{rt} + \nu_{mrt}^{nc}$$

Validating identifying assumptions

- **Key identifying assumption:** No systematic correlation of S_{bt} and municipality-level characteristics D_{mt} to y_{mt} (Exogeneity condition for our IV)

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Banks with low/high shocks to their national supply of BNDES-originated credit S_{bt} have **not systematically** sorted into **municipalities** on the basis of shocks to or differential trends in the given outcome y_{mt}

Validating identifying assumptions

- **Key identifying assumption:** No systematic correlation of S_{bt} and municipality-level characteristics D_{mt} to y_{mt} (Exogeneity condition for our IV)
- (1) Local branches of eligible banks do not arise endogenously in response to the Cartão program
 - ← Cartão loans were provided by bank branches existed in municipality in the prior year
 - ← Cartão is a minor part of most bank branches' SME portfolios

Validating identifying assumptions

- **Key identifying assumption:** No systematic correlation of S_{bt} and municipality-level characteristics D_{mt} to y_{mt} (Exogeneity condition for our IV)
- (2) Direct test by regressing $\widetilde{\text{shock}}_{m,t+1}^{\text{Cartão}}$ on lagged growth in measures of firm activity and local economic development. [▶ IV testing](#)

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Result 1 - Cartão Loans and Credit Multipliers

$$\ln(\text{non Cartão lending}) = \alpha + \gamma \ln(\text{credit}_{mrt}^{\text{carto}}) + \theta_{rt} + \varepsilon_{mrt}$$

Table 4: *Cartão* BNDES and a Credit Multiplier

	Dep. Var.: log Total credit in t excluding <i>cartão</i>			
	(1)	(2)	(3)	(4)
	OLS	IV	IV	IV
	All Municipalities		Low Initial Credit/Capita	High Initial Credit/Capita
log <i>Cartão</i> BNDES disbursements	0.251*** (0.004)	0.178*** (0.019)	0.211*** (0.012)	-0.137 (0.089)
Microregion $\times$ Year FE	Yes	Yes	Yes	Yes
Dependent Variable mean	15.9	15.9	14.1	17.6
First Stage F Statistic		134.9	127.8	31.4
Observations	55,602	55,566	27,176	27,054

Notes: Estimates of equation (18) relating log Total credit disbursements excluding *cartão* BNDES to the log of *cartão* BNDES disbursements. Columns 1 and 2 include all municipality-years. Column 3 includes municipalities with total non-BNDES credit per capita below the median in 2004, and column 4 includes municipalities above the median. The regressions include microregion $\times$ year fixed effects. Standard errors are clustered at the microregion level.

Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 2a - Entry

Table 5: Credit Supply Shocks and Local Firm Entry

Dep. Var.: log number of new firms	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	IV	OLS	IV	OLS	IV
log <i>Cartão</i> BNDES disbursements	0.181*** (0.003)	0.136*** (0.015)	0.123*** (0.002)	0.067*** (0.020)		
log non- <i>Cartão</i> BNDES disbursements			0.093*** (0.001)	0.105*** (0.015)		
log total credit disbursements					0.468*** (0.019)	0.744*** (0.022)
Microregion × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable mean	2.2	2.2	2.2	2.2	2.2	2.2
First Stage <i>F</i> Statistic		134.9		70.9		38.7
Observations	55,646	55,602	55,646	55,602	55,646	55,602

Notes: This table reports estimates for firm entry based on our three alternative estimating equations aimed at capturing local credit supply shocks. Columns 1 and 2 estimate the OLS and IV specification of equation (18), columns 3 and 4 estimate the OLS and IV specification of equation (21), and columns 5 and 6 estimate the OLS and IV specification of equation (23). The regressions include microregion × year fixed effects. Standard errors are clustered at the microregion level. The first-stage *F* statistics are based on the multivariate Kleibergen and Paap (2006) version. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 2b - Survival and Employment Growth

Table 6: Credit Supply Shocks and Local Firm Dynamics
Instrumental Variables Estimates

	log # entrants		log # of exiters		Δ log employment	
	(1)	(2)	(3)	(4)	(5)	(6)
log <i>Cartão</i> disbursements	0.136*** (0.015)		0.107*** (0.014)		-0.002 (0.003)	
log total credit disbursements		0.744*** (0.022)		0.613*** (0.021)		-0.007 (0.006)
Microregion $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable mean	2.2	2.2	2.0	2.0	0.06	0.06
First Stage <i>F</i> Statistic	134.9	38.7	134.9	38.7	134.9	38.7
Observations	55,602	55,602	55,602	55,602	55,471	55,471

Notes: This table uses the same specification as columns 2 and 6 of Table 5 in odd- and even-numbered columns respectively. The dependent variables include the log of the total number of new firms entering in year t (columns 1–2), log of the total number of firms exiting in year t (columns 3–4), and the difference in the log of total employment between year t and $t - 1$. See the notes to that table for additional details. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 3 - Changes in Entrant Capability

Table 9: Credit Supply Shocks and Entrant Capability
Instrumental Variables Estimates

	log average size of entrants		log average size of entrants surviving ≥ 2 yrs		avg. log 2 year employment growth, surviving entrants	
	(1)	(2)	(3)	(4)	(5)	(6)
log <i>Cartão</i> BNDES disbursements	0.006 (0.006)		0.020** (0.008)		0.012** (0.005)	
log total credit disbursements		0.100*** (0.019)		0.169*** (0.028)		0.037** (0.014)
Microregion $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable mean	0.9	0.9	1.3	1.3	0.29	0.29
First Stage <i>F</i> Statistic	131	33.8	110	28.5	110	28.5
Observations	51,215	51,215	43,228	43,228	43,228	43,228

Notes: This table uses the same specification as columns 2 and 6 of Table 5 in odd- and even-numbered columns respectively. The dependent variables include the log of the average size of new firms entering in year t (columns 1–2), the log of the average size of new firms entering in year t and surviving at least two years through $t + 2$ (columns 3–4), and the difference in the log of total employment between t and $t + 2$ among firms that enter in year t and survive until $t + 2$ (columns 5–6). See the notes to that table for additional details. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 4(a) - Heterogeneous effects of credit supply shocks: credit depth

**Table 11: Heterogeneous Effects by Baseline Credit Supply
Instrumental Variables Estimates**

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	log number of new firms		log number of existing firms		Δ log employment		log average size of entrants		log average size of entrants surviving ≥ 2 yrs		avg. log 2 year employment growth, surviving entrants	
Panel A: Municipalities with Below Median Non-BNDES Credit Per Capita at Baseline												
log total <i>Cartão</i> BNDES disbursements	0.141*** (0.009)		0.128*** (0.008)		-0.001 (0.004)		0.017*** (0.006)		0.035*** (0.009)		0.015*** (0.005)	
log total credit disbursements		0.640*** (0.023)		0.592*** (0.024)		-0.010 (0.012)		0.115*** (0.026)		0.168*** (0.041)		0.028 (0.022)
Dependent Variable mean	1.5	1.5	1.1	1.1	0.07	0.07	0.77	0.77	1.07	1.07	0.24	0.24
First Stage <i>F</i> Statistic	127.8	62.9	127.8	62.9	127.9	62.7	129.7	62.5	116.8	52.3	116.8	52.3
Observations	27,176	27,176	27,176	27,176	27,048	27,048	22,992	22,992	18,816	18,816	18,816	18,816
Number of Microregion $\times$ Year	4,059	4,059	4,059	4,059	4,056	4,056	3,949	3,949	3,473	3,473	3,473	3,473
Panel B: Municipalities with Above Median Non-BNDES Credit Per Capita at Baseline												
log total <i>Cartão</i> BNDES disbursements	-0.053 (0.066)		-0.050 (0.065)		0.002 (0.006)		-0.038* (0.022)		-0.038 (0.030)		0.005 (0.016)	
log total credit disbursements		0.140 (0.370)		0.169 (0.360)		-0.010 (0.044)		0.201 (0.135)		0.112 (0.124)		-0.072 (0.088)
Dependent Variable mean	3.3	3.3	3.0	3.0	0.05	0.05	1.0	1.0	1.5	1.5	0.33	0.33
First Stage <i>F</i> Statistic	31.4	1.8	31.4	1.8	31.4	1.8	29.5	2.1	22.0	2.9	22.0	2.9
Observations	27,054	27,054	27,054	27,054	27,054	27,054	26,808	26,808	23,135	23,135	23,135	23,135
Number of Microregion $\times$ Year	3,901	3,901	3,901	3,901	3,901	3,901	3,882	3,882	3,424	3,424	3,424	3,424

Notes: This table uses the same specification as columns 2 and 6 of Table 5 in odd- and even-numbered columns respectively. Across panels, we split the sample into municipalities with above versus below median non-BNDES-originated credit per capita in 2004. The dependent variables are as defined in prior tables. See the notes to Table 5 for additional details. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 4(b) - Heterogeneous effects of credit supply shocks: Young vs Old

Table 12: Heterogeneous Effects by Firm Age
Instrumental Variables Estimates

	average log 2 year employment growth among incumbents age... in t					
	1-2 years		3-10 years		11+ years	
	(1)	(2)	(3)	(4)	(5)	(6)
log <i>Cartão</i> BNDES disbursements	0.007 (0.005)		0.005* (0.003)		0.001 (0.003)	
log total credit disbursements		0.058*** (0.014)		0.028*** (0.010)		-0.006 (0.009)
Microregion $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable mean	0.163	0.163	0.111	0.111	0.131	0.131
First Stage F Statistic	118	29.5	121.4	33.4	122.2	34.5
Observations	44,285	44,285	47,396	47,396	48,891	48,891

Notes: This table uses the same specification as columns 2 and 6 of Table 5 in odd- and even-numbered columns respectively. The dependent variable measures the average two-year forward growth rate in employment among incumbent firms that have been in operation 1–2 years (columns 1–2), 3–10 years (column 3–4), and more than 11 years. See the notes to Table 5 for additional details. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Outline

- 1 Background
- 2 Model
- 3 Identification
- 4 Results
- 5 Conclusion**
- 6 Comment

Conclusion

New evidence on how credit supply expansion shape firm dynamics at the firm/local market level.

- Brazil's large-scale expansion of subsidized credit for SMEs (2002-2013): Cartão credit line
- Empirical findings:
 - ➊ Greater entry, greater exit, but no net effect on local employment
 - ➋ Increased avg. capability of entering firms (larger size, survived longer, grew faster)
 - ➌ Selection effects more pronounced in localities with weak credit markets

Outline

- 1 Background
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- 6 **Comment**

Comments

- **(1) Firm vs. Establishment level outcome**

- Authors assign each firm to its modal municipality by establishment employment.
- Crucial identifying assumption = credit supply is local in that firms in municipality m respond to changes in credit in m and not in nearby municipalities. What about Establishment across different municipalities for the same firm?
- eg) Firm using internal network to allocate resources in response to local shocks (Giroud and Muller, 2019 AER)

Comments

- **(2) More transparent ways to validate IV**

show Rotemberg weight / misspecification sensitivity,
over-identification test for instrument variables $\Rightarrow$ which bank is
driving the estimate γ and see if its correlation with spatial locations
of branches across municipalities (e.g. Goldsmith-Pinkham et al, 2020
AER)

*Rotemberg weight = covariance between the k th instrument's fitted value
of the endogenous variable and the endogenous variable itself ▶ IV construction

- Approach to follow Greenstone, Ma, Nyuen (2020, AEJ: Economic Policy): calculate the correlation between municipality i 's fixed effect (D_{mt}) and market-share weighted average fixed effect of banks located in that municipality $m \rightarrow$ close to zero correlation would suggest that banks with more shocks to the national supply are not systematically sorted into municipalities with positive shocks

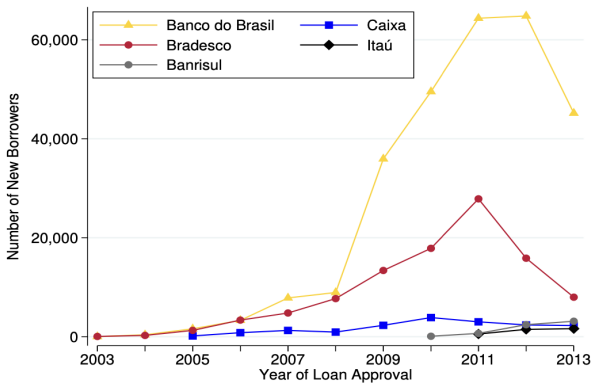
Comments

- (3) Would be more clear to see how the spillover effect worked by checking if there were **changes in loan rates for both Cartão and BNDES loan**
Also, as the model predicted, what happens to local wages in different municipalities other than employment growth?
- Limitation of the model/empirical analysis: not taking into account of cross-over effect between local labor markets and product markets. Firm's response in space dimension after credit supply expansion, product market to fully quantify the credit multiplier effect on overall employment

Description (1a)

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Figure 1: Expansion of Cartão BNDES Lending

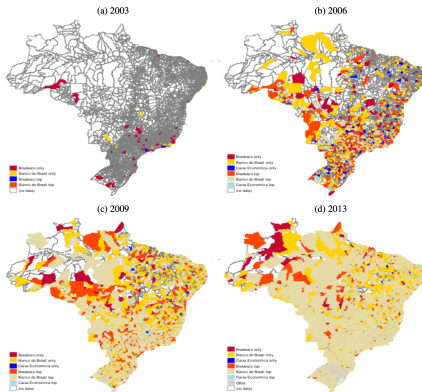


Notes: New borrowers are Cartão clients by intermediating bank by year.

Source: BNDES Indirect Finance Approvals Database.

Description (1b) [Go back](#)

Figure 2: Expansion of Cartão BNDES Lending Across Municipalities



Notes: Cartão loans by intermediating banks from 2003 to 2013. The dark colors indicate municipalities with only one of the three eligible banks—Bradesco, Banco do Brasil (BdB), or Caixa—intermediating such loans. The light colors indicate, in cases with more than one bank intermediating, which of the three has the largest outstanding Cartão loan value.

Source: BNDES Indirect Finance Approvals Database.

Description (2a)

[◀ Go back](#)**Table 1: Municipality-Level Summary Statistics**

	(1) mean	(2) median	(3) std dev	(4) observations
log total <i>Cartão</i> BNDES disbursements	6.0	7.2	6.1	55,646
log total non- <i>Cartão</i> BNDES disbursements	8.1	12.0	7.2	55,646
log total non-BNDES disbursements	15.7	15.7	2.9	55,651
log number of new firms	2.2	2.1	1.6	55,646
log number of exiting firms	2.0	1.8	1.5	55,646
log number of employees at new firms that will survive at least 2 years	1.3	1.2	0.8	43,333
log growth in employment for surviving 2 year old firms	0.29	0.29	0.49	43,333
log total formal employment	6.95	6.69	1.53	55,581
log formal employment growth	0.059	0.046	0.391	55,512

Notes: Summary statistics for core regressors and outcomes defined at the municipality–year level.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Description (2b)

[◀ Go back](#)**Table 2:** Firm-Level Summary Statistics by Credit Access

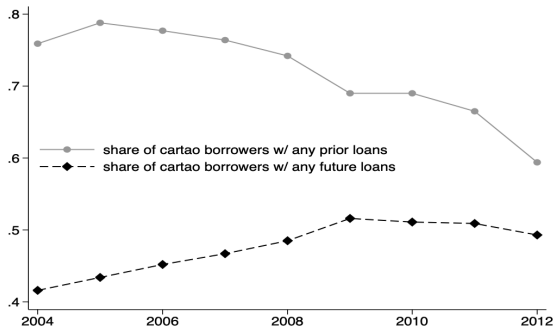
	(1) mean	(2) median	(3) std. dev.	(4) no. of firm-years
(a) All Firms				
Number of Employees	18.4	3	483.9	22,011,533
Years in Formal Operation	9.6	6	10.5	22,011,533
(b) Firms with <i>Cartão</i> BNDES Loans				
Number of Employees	24	7	96.7	693,007
Years in Formal Operation	10.6	8	9.6	693,007
(c) Firms with no <i>Cartão</i> but some other Loans				
Number of Employees	25.7	4	590.5	9,825,520
Years in Formal Operation	9.5	6	10.2	9,825,520
(d) Firms with no Loans				
Number of Employees	11.8	2	387	11,493,006
Years in Formal Operation	9.6	6	10.7	11,493,006

Notes: Summary statistics at the firm-year level.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Description (2c) [◀ Go back](#)

Figure B.1: *Cartão* BNDES, Prior and Future Borrowing



Notes: Share of all firms with *Cartão* loans that have any prior loans and, separately, any future loans.

Sources: SCR and BNDES Indirect Finance Database.

Validating Identifying assumptions (2)

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Table 3: Tests for Endogenous Targeting

Regressor	Dep. Var.: $\widehat{\text{shock}}_{cartao, t+1}$	
	β (std. err.)	No. of Obs.
GDP/capita growth, t	0.00001 (0.002)	49,968
Δ log number of new firms, t	-0.0014* (0.001)	50,058
Δ log average entry size of new firms, t	-0.0004 (0.001)	43,909
Δ log average entry size of surviving new firms, t	0.00002 (0.001)	35,571
Δ log average growth of surviving new firms, t	0.001 (0.001)	35,571
Δ log number of exiting firms, t	0.00003 (0.001)	50,058
Δ log employment growth, t	-0.0008* (0.000)	49,917
Δ log employment growth among new firms, t	-0.002 (0.002)	49,917
Δ log employment growth among new firms, t	-0.002 (0.002)	49,917
Δ log number of bank branches, t	0.004 (0.005)	50,062

Notes: Regressions of instrument, defined in equation (20), on the lagged difference in our core outcomes of interest at the municipality level. Each row is a separate regression, each of which controls for microregion $\times$ year fixed effects. The sample size varies across regressions as the outcomes are only defined for municipalities with the given set of firms in the given adjacent years. Standard errors are clustered at the microregion level.

Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Validating Identifying assumptions (1)

◀ Go back

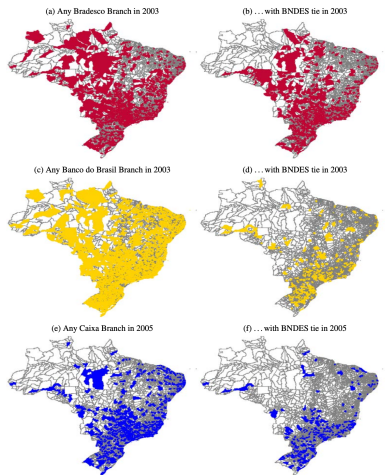


Figure 3: Location of Cartão-Eligible Bank Branches

Validating Identifying assumptions (2)

[◀ Go back](#)

Table 3: Tests for Endogenous Targeting

Regressor	Dep. Var.: $\widehat{\text{shock}}_{cartao, t+1}$	
	β (std. err.)	No. of Obs.
GDP/capita growth, t	0.00001 (0.002)	49,968
Δ log number of new firms, t	-0.0014* (0.001)	50,058
Δ log average entry size of new firms, t	-0.0004 (0.001)	43,909
Δ log average entry size of surviving new firms, t	0.00002 (0.001)	35,571
Δ log average growth of surviving new firms, t	0.001 (0.001)	35,571
Δ log number of exiting firms, t	0.00003 (0.001)	50,058
Δ log employment growth, t	-0.0008* (0.000)	49,917
Δ log employment growth among new firms, t	-0.002 (0.002)	49,917
Δ log employment growth among new firms, t	-0.002 (0.002)	49,917
Δ log number of bank branches, t	0.004 (0.005)	50,062

Notes: Regressions of instrument, defined in equation (20), on the lagged difference in our core outcomes of interest at the municipality level. Each row is a separate regression, each of which controls for microregion $\times$ year fixed effects. The sample size varies across regressions as the outcomes are only defined for municipalities with the given set of firms in the given adjacent years. Standard errors are clustered at the microregion level.

Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Result 2c - Survival and Employment Growth

$$y_{imt} = \alpha + \gamma \ln(\text{credit}_{imrt}^{\text{carto}}) + \delta \ln(\text{credit}_{imt}^{\text{bndes}}) + \theta_i + \theta_t + \varepsilon_{imt}$$

Table 8: Firm-Level Effects
Instrumental Variables Estimates

	exit in t		Δ log employment	
	(1)	(2)	(3)	(4)
log own-firm <i>Cartão</i> BNDES credit	-0.067 (0.077)	-0.079 (0.081)	0.255 (0.203)	0.236 (0.168)
log own-firm non- <i>Cartão</i> BNDES disbursements		0.013 (0.037)		0.023 (0.066)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Dependent Variable mean	0.030	0.030	0.082	0.082
First Stage F Statistic	1.0	0.7	1.2	0.8
Observations	2,420,120	2,420,120	2,188,110	2,188,110

Notes: This table reports results based on a firm-level panel IV regression. The dependent variables include an indicator for whether the firm exits in year t (columns 1–2) and the difference in the log of total firm employment between year t and $t - 1$. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, and BNDES Indirect Finance Database.

Robustness checks

Table 7: Identification Checks on Table 6

	log # entrants		log # of exiters		Δ log employment	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Year FE $\times$ Predetermined GDP/capita and Manuf. Shr.						
log <i>Cartão</i> disbursements	0.132*** (0.015)		0.103*** (0.014)		-0.002 (0.003)	
log total credit disbursements		0.747*** (0.024)		0.610*** (0.023)		-0.007 (0.007)
First Stage <i>F</i> Statistic	143.9	40.9	143.9	40.9	143.9	40.9
Panel B: Twice-Lagged Bank-Specific <i>Cartão</i> Share						
log <i>Cartão</i> disbursements	0.096*** (0.019)		0.071*** (0.014)		-0.002 (0.004)	
log total credit disbursements		0.784*** (0.024)		0.600*** (0.042)		-0.012 (0.013)
First Stage <i>F</i> Statistic	132.1	20.3	132.1	20.3	132.1	20.3
Panel C: Year FE $\times$ Predetermined Outcome Variable						
log <i>Cartão</i> disbursements	0.031*** (0.009)		0.042*** (0.008)		-0.002 (0.002)	
log total credit disbursements		0.560*** (0.102)		0.387*** (0.050)		-0.005 (0.006)
First Stage <i>F</i> Statistic	164	10.9	187.4	30.1	134.9	38.1
Microregion $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Dependent Variable mean	2.2	2.2	2.0	2.0	0.06	0.06
Observations	55,602	55,602	55,602	55,602	55,471	55,471

Notes: This table re-estimates Table 6 with different sets of controls. Panel A includes year fixed effects interacted with municipality-level GDP/capita and manufacturing share of employment in 2003, the initial year of the panel prior to the rollout of the *Cartão* BNDES program. Panel B includes the twice-lagged share of total *Cartão* in the municipality by each of the seven banks eligible for intermediating the credit at that point in time. Panel C includes year fixed effects interacted with the given column's municipality-level outcome variables in 2003, the initial year of the panel prior to the rollout of the *Cartão* BNDES program. See the notes to Table 6 for additional details. The three rows at the bottom of the table applies to all three panels. Standard errors clustered by microregion. Significance levels: * : 10% ** : 5% *** : 1%.

Sources: RAIS, SCR, IBGE, and BNDES Indirect Finance Database.